

NOVOSELENGINSK, Russian Federation

WMO: 308290

Lat: 51.102N

Lon: 106.642E

Elev: 556

StdP: 94.82

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.9	-33.7	-38.5	0.1	-35.8	-36.4	0.1	-33.5	8.4	-12.8	7.1	-14.8	0.7	270	0.637

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	31.3	18.0	29.3	17.5	27.4	16.9	19.8	27.7	18.9	26.3	18.1	25.0	3.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.4	13.3	22.4	16.4	12.5	21.7	15.5	11.8	21.2	59.3	28.0	56.3	26.3	53.7	25.1	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.5	9.1	7.6	-38.0	35.0	2.8	2.5	-40.0	36.8	-41.6	38.3	-43.2	39.7	-45.2	41.6
			-38.1	21.5	2.8	1.0	-40.1	22.2	-41.7	22.8	-43.3	23.4	-45.3	24.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.1	-24.3	-19.6	-7.2	3.7	10.8	18.1	20.9	18.1	10.4	1.1	-10.7	-21.0	(6)
(7)		DBStd	16.38	5.36	5.73	6.87	4.55	4.49	3.35	3.30	4.10	4.45	6.49	6.06	(7)	
(8)		HDD10.0	4568	1062	829	533	194	44	1	0	45	276	622	962	(8)	
(9)		HDD18.3	6826	1320	1063	792	438	236	53	12	46	241	533	872	1220	(9)
(10)		CDD10.0	966	0	0	0	5	71	245	339	250	56	1	0	0	(10)
(11)		CDD18.3	183	0	0	0	0	3	48	93	38	1	0	0	0	(11)
(12)		CDH23.3	1811	0	0	0	3	81	538	839	329	20	0	0	0	(12)
(13)		CDH26.7	576	0	0	0	0	20	178	289	86	3	0	0	0	(13)
(14)	Wind	WSAvg	2.6	1.4	1.5	2.6	3.6	4.0	3.3	2.8	2.6	2.9	2.5	2.0	1.6	(14)
(15)	Precipitation	PrecAvg	222	4	1	1	6	15	36	58	59	25	6	6	6	(15)
(16)		PrecMax	299	12	4	4	15	35	86	107	110	48	15	12	12	(16)
(17)		PrecMin	161	1	0	0	1	4	8	25	24	7	0	1	1	(17)
(18)		PrecStd	41	2	1	1	5	8	18	26	21	11	4	3	3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.5	-0.6	13.4	22.1	29.7	33.6	34.9	32.1	25.9	16.5	6.0	-3.5	(19)
(20)		0.4%	MCWB	-9.1	-3.4	5.3	10.8	15.5	17.9	19.6	18.8	15.9	8.7	1.6	-5.0	(20)
(21)		2%	DB	-11.4	-5.7	8.3	18.2	25.7	30.7	32.0	28.9	22.4	13.0	2.8	-7.6	(21)
(22)		2%	MCWB	-12.4	-7.6	2.5	8.4	13.3	16.7	18.9	17.8	13.7	6.2	-0.5	-8.8	(22)
(23)		5%	DB	-14.1	-8.4	5.6	15.3	23.0	28.6	29.8	26.8	19.9	10.3	0.4	-10.3	(23)
(24)		5%	MCWB	-14.9	-9.7	0.9	6.9	11.7	16.3	18.5	17.4	12.5	4.9	-2.1	-11.3	(24)
(25)		10%	DB	-16.4	-10.9	2.9	12.5	20.2	26.4	27.8	24.7	17.6	8.0	-1.7	-12.5	(25)
(26)		10%	MCWB	-17.1	-11.9	-0.8	5.3	10.4	15.5	17.7	16.8	11.2	3.7	-3.6	-13.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.9	-3.7	5.5	11.0	15.6	19.7	21.4	20.2	16.4	9.0	2.0	-4.9	(27)
(28)		0.4%	MCDWB	-7.5	-0.7	12.9	21.8	28.0	29.4	29.9	27.7	24.5	15.7	5.5	-3.4	(28)
(29)		2%	WB	-12.4	-7.5	2.8	8.8	13.8	18.3	20.2	19.0	14.5	6.7	-0.2	-8.7	(29)
(30)		2%	MCDWB	-11.5	-5.7	7.9	17.3	24.9	27.1	28.3	26.4	20.8	11.9	2.5	-7.6	(30)
(31)		5%	WB	-14.9	-9.8	1.0	7.1	12.2	17.3	19.5	18.2	13.2	5.2	-2.0	-11.2	(31)
(32)		5%	MCDWB	-14.1	-8.5	5.4	14.9	21.5	25.7	27.1	24.8	18.6	9.8	0.3	-10.3	(32)
(33)		10%	WB	-17.1	-11.9	-0.8	5.5	10.9	16.4	18.7	17.4	11.8	3.9	-3.6	-13.2	(33)
(34)		10%	MCDWB	-16.5	-10.9	2.8	12.2	19.3	24.6	25.7	23.3	16.9	7.8	-1.7	-12.4	(34)
(35)	Mean Daily Temperature Range	MDBR	9.0	12.6	13.5	12.7	12.9	11.7	10.3	10.5	11.2	11.2	9.6	8.3	(35)	
(36)		5% DB	MCDBR	10.3	14.1	15.4	18.0	18.7	15.8	13.9	14.5	15.0	15.0	11.3	9.3	(36)
(37)			MCWBR	9.5	12.3	9.9	10.5	9.3	6.2	4.7	5.6	7.3	8.9	8.4	8.5	(37)
(38)			MCDBR	10.2	14.0	14.8	17.1	16.6	13.2	10.9	12.2	13.1	14.2	10.9	9.3	(38)
(39)		5% WB	MCWBR	9.4	12.2	9.5	10.2	8.5	5.6	4.3	5.3	6.6	8.6	8.3	8.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.269	0.293	0.330	0.378	0.443	0.438	0.429	0.390	0.337	0.310	0.290	0.270	(40)	
(41)		taud	2.197	2.181	2.213	2.217	2.050	2.186	2.257	2.351	2.447	2.436	2.289	2.176	(41)	
(42)		Ebn at Noon	744	827	862	859	813	821	822	839	854	810	715	670	(42)	
(43)		Edh at Noon	71	96	113	127	158	139	128	111	90	74	65	62	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.41	2.52	3.92	4.99	5.97	6.17	5.71	4.84	3.77	2.40	1.41	1.02	(44)	
(45)		RadStd	0.06	0.10	0.19	0.28	0.31	0.32	0.33	0.26	0.33	0.16	0.08	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page